

Figure 1 is a plot showing the Dispersion Energy (kcal/mol) versus the number of basis functions (N) for the SAPT2+3 method. The plot compares the results of the SAPT2+3(CCD)/aTZ method (black line with circles) with the SAPT(PBE0)-D4 method (various lines and markers) and the Equilibrium curve (dashed grey line). The x-axis represents the number of basis functions (N) from 1.5 to 4.5, and the y-axis represents the Dispersion Energy (kcal/mol) from -10 to 0. The SAPT2+3(CCD)/aTZ method shows the lowest dispersion energy, while the Equilibrium curve shows the highest. The SAPT(PBE0)-D4 method results are very close to each other and to the SAPT2+3(CCD)/aTZ method.

N	Equilibrium	SAPT(PBE0)-D4 (S, ND)	SAPT(PBE0)-D4 (I, ND)	SAPT(PBE0)-D4 (S)	SAPT(PBE0)-D4 (I)	SAPT2+3(CCD)/aTZ
1.5	-7.5	-9.2	-9.2	-7.2	-7.5	-7.2
2.0	-5.5	-5.5	-5.5	-5.2	-5.5	-5.2
2.5	-3.5	-3.5	-3.5	-3.2	-3.5	-3.2
3.0	-2.5	-2.5	-2.5	-2.2	-2.5	-2.2
3.5	-1.8	-1.8	-1.8	-1.5	-1.8	-1.5
4.0	-1.2	-1.2	-1.2	-1.0	-1.2	-1.0
4.5	-0.8	-0.8	-0.8	-0.6	-0.8	-0.6

Figure 1 is a line graph showing the dispersion energy (in kcal/mol) of the H₂O...H₂O dimer as a function of distance (in Å). The y-axis ranges from -10 to 0 kcal/mol, and the x-axis ranges from 1.5 to 4.2 Å. A vertical dashed line is drawn at 2.0 Å. The graph compares six theoretical curves:

- SAPT(PBE0)-D4 (S): Red line with square markers.
- SAPT(PBE0)-D4 (I): Green line with triangle markers.
- HF-D4: Blue line with circle markers.
- SAPT(PBE0)/aDZ: Brown line with cross markers.
- SAPT(PBE0)/aTZ: Grey line with plus markers.
- SAPT2+3(CCD)/aTZ: Black line with circle markers.

The curves show that the dispersion energy increases (becomes less negative) as the distance increases. The SAPT(PBE0)-D4 (S) and SAPT(PBE0)-D4 (I) curves are the most accurate, closely following the SAPT2+3(CCD)/aTZ curve. The HF-D4 curve is the least accurate, showing a much steeper increase in energy at larger distances.